

1(a)(i)	plasmodesmata ; I cytoplasmic strands / cytoplasm R if other structures also given	1
1(a)(ii)	(existence of / down, a) water potential gradient ; I <i>ref. to</i> solute / osmotic, potential A lower / more negative, water potential of (cell) C (than cell B) A higher / less negative, water potential of (cell) B (than cell C) A Ψ for water potential	1
1(a)(iii)	symplastic (pathway) ; I vacuolar / vacuole, pathway A symplast	1
1(b)	<i>max 3 if no ref. to evaporation or water vapour and ref. only to water</i> <i>any four from:</i> 1 evaporation of water / water to water vapour ; <i>must be in context of P</i> 2 P described as, surface / cell wall, of (spongy) mesophyll (cell) ; R palisade mesophyll cell A film / layer, of moisture / water (for surface of cell) 3 water vapour enters (substomatal) <u>air space</u> ; R <u>intracellular</u> air space A other correct refs. to air space in sequence of events 4 <u>diffusion</u> of water vapour (out) through stoma / stomatal pore (to Q) ; R through guard cells R diffusion by osmosis 5 (towards Q because) down <u>water potential gradient</u> / from high to low <u>water potential</u> / to lower <u>water potential</u> ; I potential gradient alone A down (water) vapour pressure gradient A Ψ for water potential <i>context is movement out to atmosphere, <u>not</u> movement from P to air space</i> 6 AVP ; e.g. stoma stated as open heat energy required for evaporation / <i>ref. to</i> latent heat of vapourisation air space described as, intercellular / saturated / substomatal	4

Question	Answer	Marks
1(c)	1 correct formula ; (magnification) = image length ÷ actual length ; A rearranged formula A magnification triangle 2 correct working ; 3 correct answer for magnification ; R if units given (×) 1440 (36 mm) / 1460 (36.5 mm) / 1480 (37 mm) / 1500 (37.5 mm) 1520 (38 mm) / 1540 (38.5 mm) / 1560 (39 mm) ; <i>if answer given to 2 sig. figs, look for the correct answer at the end of working or allow if stated by answer that this is to 2 sig. figs.</i> <i>allow ecf for incorrect conversions used in working or incorrect measured value of X–Y</i>	3

